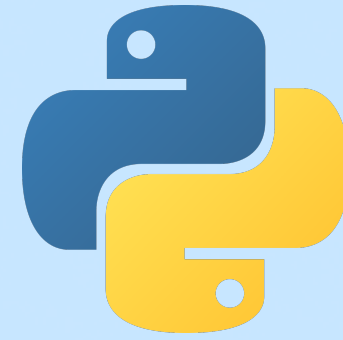


Introduction and basics

Command line, Python, Jupyter

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Contenu du cours

1. Manipulation de la ligne de commande via terminal.

- a. Chemins
- b. Commandes Bash
- c. Editeur de text VIM

2. Python

- 1. Pip et venv

[New] 2. Uv

[New] 3. Outils de Coding avec IA dans VS Code

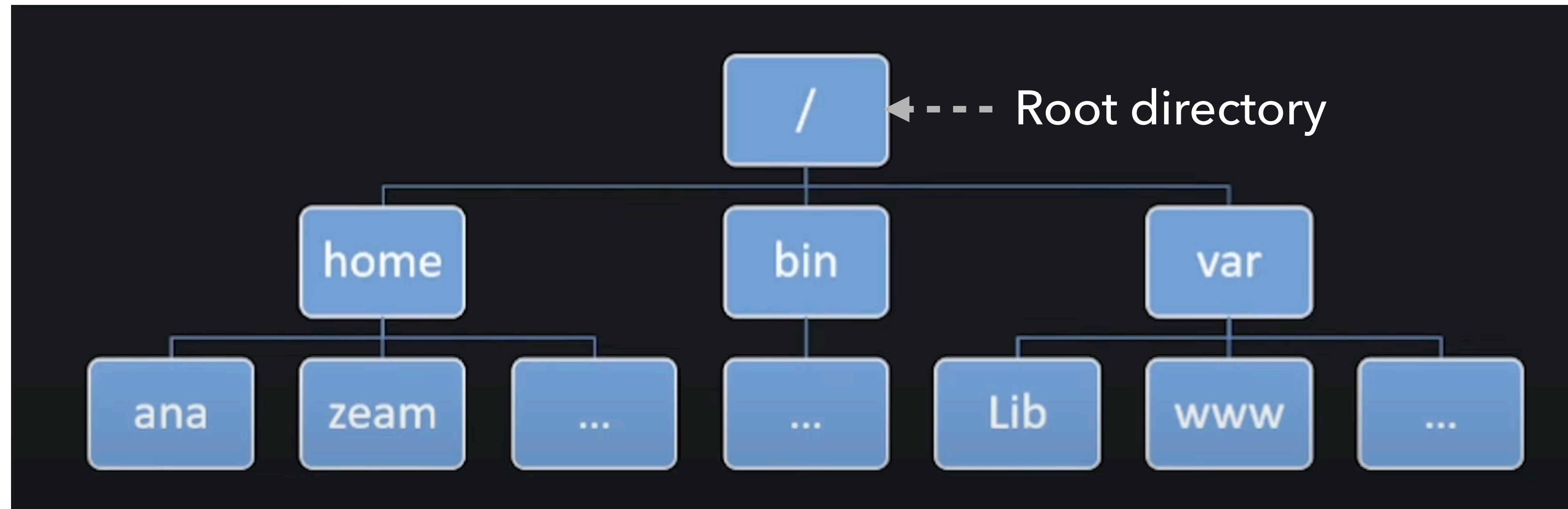
3. Jupyter et *voilà*

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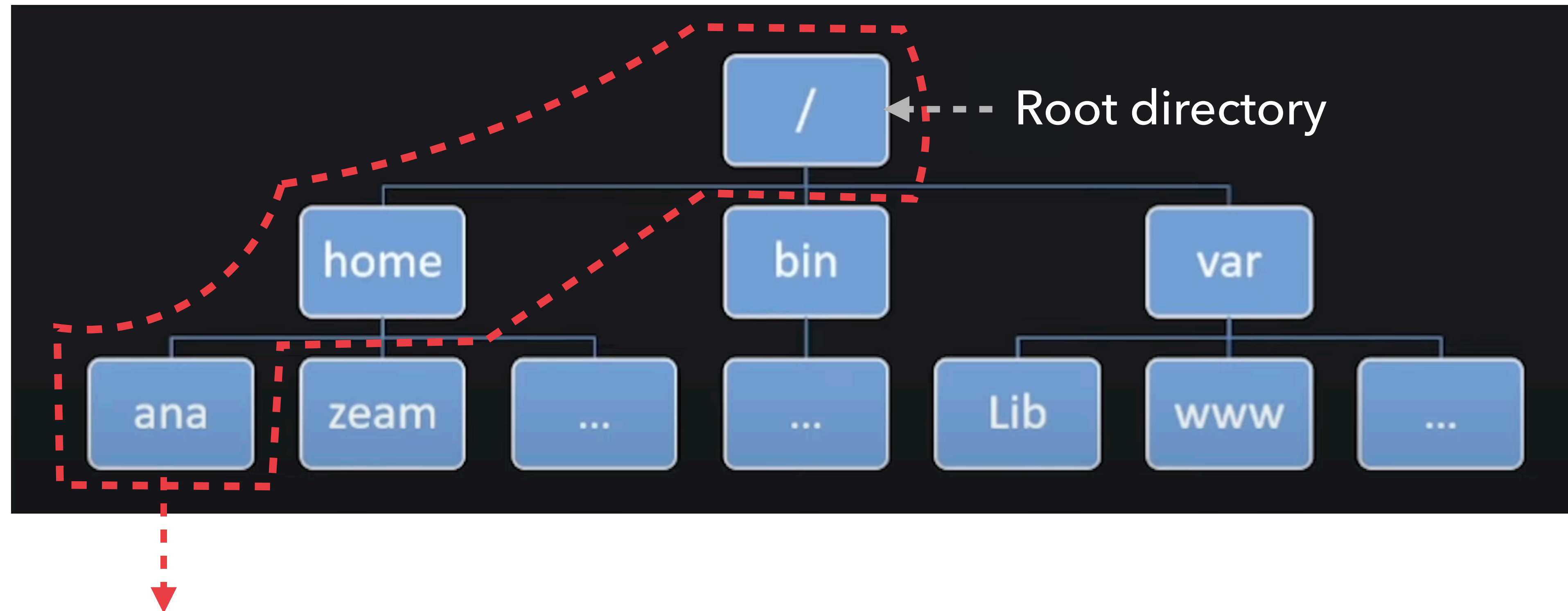
Linux

Command line

Linux file system

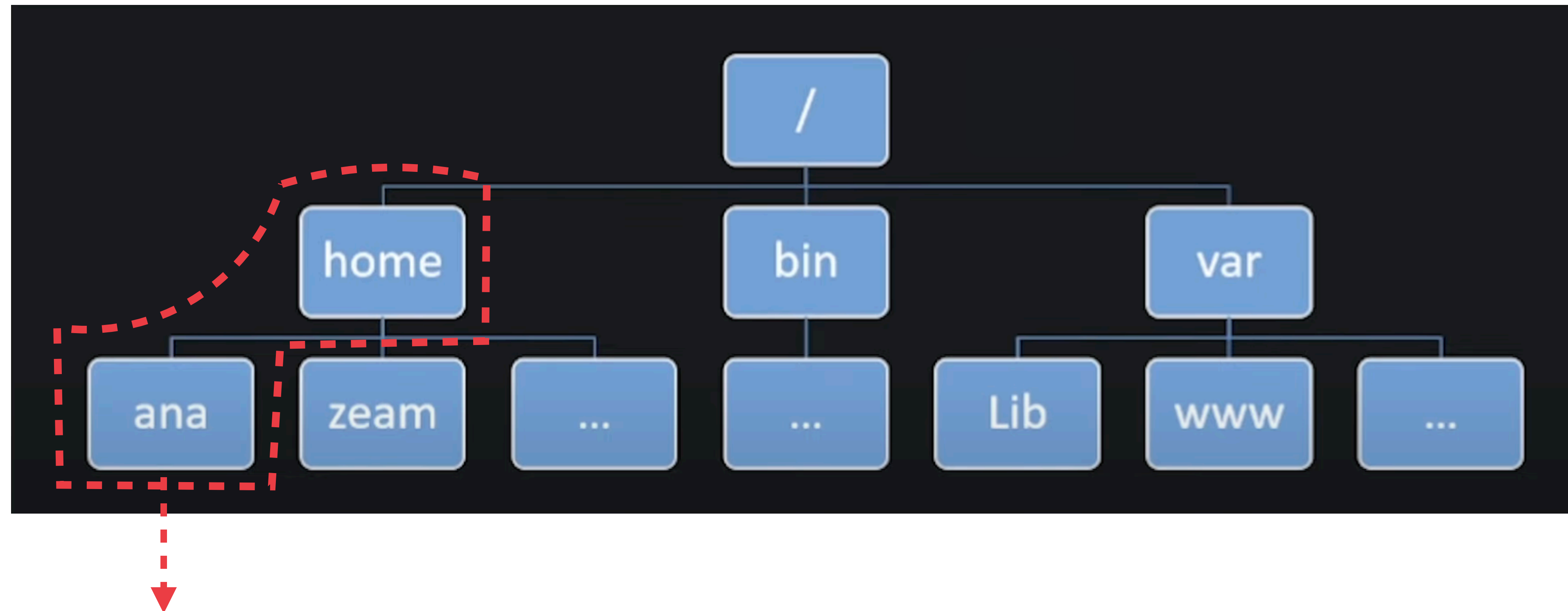


Linux file system



Absolute path **/home/ana** (Full path from root)

Linux file system



Relative path (Depends on your current directory)

Command line basics

```
$ pwd    #Print working directory
```

```
$ cd <dir> #Change directory
```

```
$ cd ..  #Change to parent directory
```

```
$ ls -A  #List
```

```
$ mkdir <dir(s)> #Create one or multiple directories
```

```
$ touch <file(s)> #Create one or multiple files
```

Command line basics

\$ pwd #Print working directory

\$ cd <dir> #Change directory

\$ cd .. #Change to parent directory

\$ ls -A #List

\$ mkdir <dir(s)> #Create one or multiple directories

\$ touch <file(s)> #Create one or multiple files

\$ cp <src> <dest> #Copy file

\$ cp -r <src> <dest> #Copy directory

\$ mv <src> <dest> #Move/rename file

\$ rm <file> #Remove file

\$ rm *.txt #Remove all text files

\$ rm -r <dir> #Remove directory

Command line basics

```
$ history #history > hist.txt
```

```
$ clear
```

```
$ echo
```

```
$ chmod #Change file permissions
```

```
$ grep #history | grep cd
```

```
$ diff #Compare two text files line by line
```

Command line basics

```
$ man <command> #Manual page
$ locate <file> #Locate a file in the system
$ cat <file> #Display a file in the terminal
$ open . #(or start .) display the current folder
$ head #Show the first lines of a file
$ tail #Show the last lines of a file
$ exit #Exit the current shell/terminal session
$ ps # (Process Status) Show the list of running processes
$ kill #Terminate a process by its Process ID (PID)
```

```
$ history #history > hist.txt
$ clear
$ echo
$ chmod #Change file permissions
$ grep #history | grep cd
$ diff #Compare two text files line by line
```

1. Create your workspace

- Go to your **home directory** (`pwd` to check).
- Create a folder named `M1_project` (`mkdir`).
- Inside it, create 3 subfolders:
 - `data/` → for datasets
 - `src/` → for Python scripts
 - `notebooks/` → for Jupyter notebooks

2. Add some files

- Inside `src/` create two files: `main.py` and `utils.py` (`touch`).
- In `data/` create two files: `train.csv` and `test.csv`.
- Verify that everything is created (`ls -A`).

3. Organize the files

- Move `main.py` into the root folder `M1_project/` (`mv`).
- Copy `utils.py` into `notebooks/` (`cp`).
- Rename `train.csv` into `training.csv`.

4. Clean up mistakes

- You don't need `test.csv`. Delete it (`rm`).
- Also, remove the `utils.py` inside `notebooks/` using `rm`.

5. Explore history and filters

- Show the list of your previous commands (`history`).
- Save them into a file called `history.txt`.
- From this file, extract only the commands where you used `cd` (`grep`).

Use VIM

VI (**V**isual editor) and VIM (**V**i **I**Mproved) are default text editor on almost all Unix/Linux systems.

```
$ vim main.py
```

The modes are shown at the bottom left of the editor

- The **command mode** is the default mode.
- To go back to command mode type **esc** key.
- To start modifying (**Insert mode**) type **i** key on keyboard.
- To **exit**, go back to command mode and
 - no modifications, type **:q**
 - modified file
 - save modifications **:wq**
 - exit without save **:q!**